Lab 6: Simulation Studies

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Our lab today focuses on section **3. Simulation studies** of the paper by Li and Lin (2025), which you have read in advance. Specifically, we'll discuss the goals of the simulation study, the choices the authors made in designing their data generating mechanisms, how those choices may affect the results of the study, and interpretation of those results as presented in Tables 2 and 3. # Schedule 1. First, to get us all refreshed on the context, I'll give a **brief overview** of their method. 2. Please form groups of 2-3 students. **Open a shared Google Doc**, giving access to everyone in your group. 3. List what were the main goals of the simulation study performed by the authors and what metrics they considered in assessing their method. - How do Emp SE and Est SE in Tables 2 and 3 differ? How large a discrepancy would concern you, and what additional information would you need to judge it? Would the average estimated SE reveal how much the estimated SEs vary across datasets? - How many simulation replications were used? If true coverage were 95%, how much Monte Carlo variability would you expect in the reported coverage? Use this to assess a reported value. Do the authors quantify this simulation uncertainty? 4. Discuss how the authors generate their **simulated datasets**, as described in Section 3.1 of the paper. In your Google Doc, have one group member write down **pseudo-code** to generate a single dataset. Identify what is generated once and what is regenerated in each replication, and state any details you need to infer. - Which quantities are independent, and conditional on what? How does your simulation preserve dependence among individual outcomes and between county totals and state demographic counts? - The simulations use independent normal county random effects. If you instead wanted correlated county effects with a specified positive-definite covariance matrix, how would you generate them efficiently using the Cholesky decomposition described in Unit 9? 5. Next, **discuss the specific choices** the authors made in designing their simulated datasets. As you discuss, **write down those choices** in your Google Doc. 6. As a group, pick the **two most important choices** and justify why you picked them in the Google Doc. 7. With those choices in mind, brainstorm and write how you might simulate the data differently. Describe a data generating process that would challenge an assumption or approximation of the fitted model, and predict how it might affect coverage or standard error estimates. 8. Finally, have one group member export the Google Doc to PDF and submit it **as a group** for the "Lab 6" assignment on **Pensive**. Only one submission is needed per group. 9. Time allowing, after your group submits its assignment to Pensive, find another group that

has also completed its submission and discuss your answers. ## Acknowledgements This lab was originally developed by Andrew Vaughn and James Duncan.